

ARASH VASHAGH

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Summary

Machine Learning Developer with research publications and industry certifications in AI, cybersecurity, and cloud. Experienced in building robust ML pipelines and adversarially resilient models using PyTorch, with hands-on experience in security tools and cloud platforms (AWS and Azure). Strong at translating research into practical solutions.

Skills

Programming: Python, Java, JavaFX, Kotlin, SQL, C++

Backend & APIs: REST API integration, JSON processing

Frameworks & Libraries: PyTorch, TensorFlow, Flask, FastAPI, Scikit-learn

Cloud Platforms: AWS, Microsoft Azure

Tools & Environments: Jupyter, Git, VS Code, IntelliJ IDEA, Docker, Wireshark

Languages: Persian (Native), English (Professional)

Certifications

- AWS Certified AI Practitioner; AWS Certified Cloud Practitioner
- CompTIA Security+; CompTIA Data+; CompTIA Project Management Essentials
- Microsoft AI-900, DP-900, AZ-900, SC-900
- DeepLearning.AI - Agentic AI
- ISC2 CC in progress; CompTIA DataSys+ beta results pending

Experience

Team Lead — Trustworthy and Secure AI (TSAI) Lab — Fredericton, Canada **Oct 2025 – Present**

- Overseeing lab operations and coordinating graduate student work.
- Reporting progress and operational updates to the supervising faculty member.
- Managing the [TSAI LinkedIn page](#) for lab communications and educational content.

Graduate Research Assistant — University of New Brunswick (UNB) — Fredericton, Canada **Sep 2024 – Present**

- MSc thesis research in Adversarial Machine Learning (AML).
- Co-authored [a survey](#) on adversarial attacks across ML and XAI.
- Co-authored a survey on defense mechanisms in adversarial machine learning (under review / currently private).
- Proposed three novel defense mechanisms (under review / currently private).
- Developing a novel adversarial attack against fusion models (under investigation / currently private).
- Multiple manuscripts under review at top journals and conferences.

Graduate Teaching Assistant — UNB — Fredericton, Canada **Jan 2025 – Apr 2026**

- *Software Engineering* (Winter 2026) – Java
- *Data Communication and Network Modeling* (Fall 2025) – Python
- *Data Science for Big Data Analytics* (Winter 2025) – Python

Conference Reviewer **Jul 2024 – Oct 2024**

- Reviewed manuscripts and provided structured feedback at international conferences:
 - *ArtInHCI 2024* – Artificial Intelligence and Human-Computer Interaction (Kunming, China) [\[Certificate\]](#)
 - *BIBE 2024* – Biological Information and Biomedical Engineering (Hohhot, China) [\[Certificate\]](#)

Research Assistant — Isfahan University of Technology (IUT) — Isfahan, Iran **Feb 2021 – Feb 2024**

- Conducted research on machine learning and healthcare, resulting in two IEEE publications [\[1\]](#) and [\[2\]](#).
- Collaborated with faculty on model design, data preprocessing, and experimental evaluation.
- Supported technical writing and presentation preparation for joint research projects.

Teaching Assistant — IUT — Isfahan, Iran **Feb 2021 – Feb 2024**

- Mentored over 400 students in undergraduate courses including Artificial Intelligence, Embedded Systems, Data Structures, and Programming Labs in C, C++, Java, and Python.
- Led tutorials, graded assignments, and guided students through programming and algorithmic exercises.

Selected Projects

PromptShield: Adversarial Detection for Vision-Language Models

Jan 2026 – May 2026

- Developed a novel framework to detect adversarial images in vision-language models.
- Evaluated the framework across multiple datasets, model backbones, and adversarial attacks.
- Technologies: Python, PyTorch, OpenCLIP, NumPy, Scikit-learn.
- Manuscript and code under review at [NeurIPS 2026](#).
- Source code: [GitHub](#).

AttackBench: Adversarial Attack Visualization Tool

Mar 2026 – Apr 2026

- Developed a desktop application to generate and visualize adversarial examples on image classification models.
- Implemented common adversarial attacks and displayed their impact on model predictions and confidence.
- Built a Java-based UI with a Python backend to load models, run attacks, and return results in real time.
- Technologies: Java (JavaFX), Python, PyTorch, FastAPI.
- Source code: [GitHub](#).

PhishGuard: Phishing URL Detection App

Mar 2026 – Apr 2026

- Developed an Android application using Kotlin with a backend service for phishing URL detection.
- Integrated RESTful API communication between mobile client and Python backend.
- Implemented input validation and security checks including URL formatting, DNS lookup, and SSL verification.
- Provides predictions with confidence scores and user-friendly feedback.
- Technologies: Kotlin, Python, Flask, Scikit-learn, Retrofit.
- Source code: [GitHub](#).

Education

Master's Candidate — University of New Brunswick (UNB), Fredericton, Canada

Sep 2024 – Aug 2026

- Awarded Graduate Scholarship and Research Grant.
- Major: Computer Science
- Research Focus: Adversarial Machine Learning

Bachelor of Science — Isfahan University of Technology (IUT), Isfahan, Iran

Sep 2019 – Feb 2024

- Major: Computer Engineering
- Minor: Intelligent Systems

Voluntary Educator (AI Security)

AI Security and Adversarial Machine Learning Course — Online

Jun 2026 – Present

- Creating a Persian technical course on AI Security and Adversarial Machine Learning to support Persian-speaking learners facing language barriers.
- Developed slides and scripts on machine learning foundations for AML, including classification, loss functions, gradients, and gradient descent.
- Published course videos through a structured [YouTube playlist](#) and shared course materials on [GitHub](#).
- Targeting completion of the Persian course by the end of June 2026, with an English version planned afterward.